Q1. The table shows some of the chemical reactions of four elements.

element	reaction with water	reaction with dilute hydrochloric acid	reaction when heated in air
A	none	none	burns to give an acidic oxide
B	vigorous, hydrogen given off	violent, gas given off	burns to give a basic oxide
C	none	none	reacts slowly to give a basic oxide
D	none	vigorous, gas given off	burns to give a basic oxide

(a) Elements B, C and D are all metals. Write the letters down in order of the reactivity of the metals starting with the **most** reactive.

most reactive ....

1 mark

(b) (i) Name **one** gas, present in the air, with which all four elements will react.

.....

1 mark

(ii) Name the gas which is produced when element D reacts with dilute hydrochloric acid.

.....

1 mark

(c) What evidence is there to show that element A is the only non-metal?

.....

.....

1 mark

(d) Give the name of an element which could be element B.

.....

1 mark

Maximum 5 marks

Mark schemes

Q1.

(a) B D C

all three letters in the correct order are required for the mark

1 (L6)

(b) (i) oxygen

1 (L6)

(ii) hydrogen

1 (L7)

(c) it's the only one which forms an acidic oxide

accept 'it has an acidic oxide'
do **not** accept 'it burns to give an acid'

1 (L7)

(d) any **one** from

- sodium
- potassium
- lithium
- barium

*accept 'caesium' **or** 'rubidium' **or** 'calcium'*
*do **not** accept 'aluminium 'or 'magnesium'*

1 (L7)

[5]